

CS2 Week 9

Convex Optimization



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Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Notes available on Kissan's website: <https://kissanv.github.io>
& on Haixiao's Polybox: <https://polybox.ethz.ch/index.php/s/AReA9YBeJQKxBDn>
- Quizzes, Examples & Old exam questions



Kissan's Website



Haixiao's Polybox

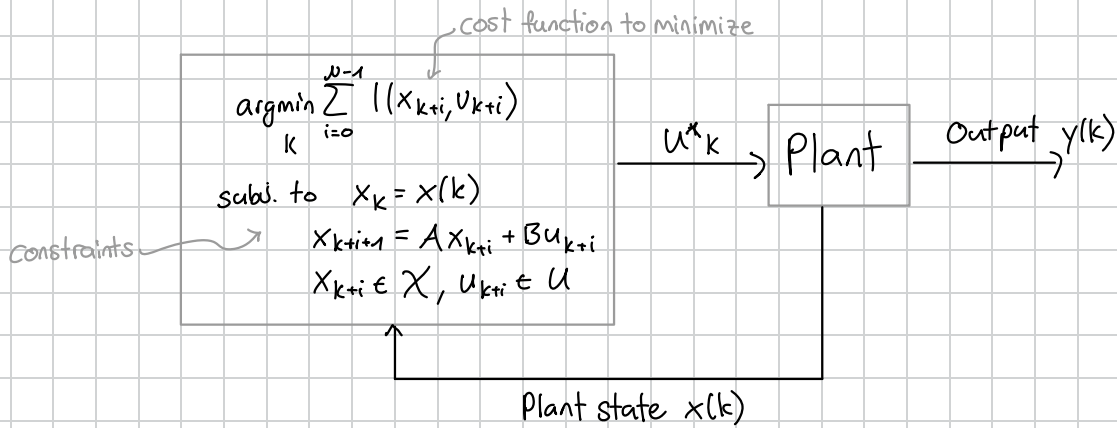
Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback, Pole Placement, LQR	3
State Estimation, Luenberger Obs., Separation Principle, LQG	4
DOFC with integral action, Intro to MIMO systems	5
Norms	6
Nonlinear Systems, Lyapunov, Feedback Linearization	7
break	-
Discrete-time Unconstrained Optimal Control	8
Convex Optimization	9
Model Predictive Control	10
Practical Model Predictive Control	11
Robust Model Predictive Control	12
Implementation Methods	13

Why do we need convex optimization?

Im trying out a different style...

Take a closer look at the mathematical formulation of MPC:



At each sample time:

- First, measure/estimate state $x(k)$
- Then, find **optimal input sequence** for entire planning window N : $U^*_k = \{u^*_k, u^*_{k+1}, \dots, u^*_{k+N-1}\}$
- Only implement first control action u^*_k

The mathematical complexity to solve this optimization problem depends entirely on the system type:

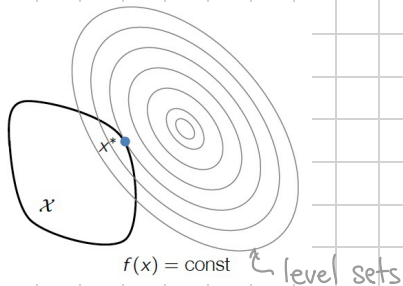
Linear Systems	Nonlinear Systems
• Linear system dynamics • Continuous set of states and inputs, e.g., $x \in [x_{\min}, x_{\max}], u \in [u_{\min}, u_{\max}]$ • Example: Chemical processes	• Nonlinear system dynamics • Continuous set of states and inputs, e.g., $x \in [x_{\min}, x_{\max}], u \in [u_{\min}, u_{\max}]$ • Example: Drones, Cars, Kites
Hybrid Systems	Discrete Decision Variables
• Mixed dynamics that are both continuous and discrete, e.g. $\begin{cases} x_{k+1} = -c_1 & x_k \geq x_{\max} \\ x_{k+1} = c_2 - c_1 & x_k < x_{\max} \end{cases}$ • Continuous set of states and inputs • Example: Walking robot	• Inputs and/or states can only take discrete values, e.g. $u \in \{1, 2, 3, 4, 5\}$ • Example: Internet

Type of system dictates solver you need to use...

Main Concepts

A mathematical optimization problem is generally formulated as:

$$\begin{aligned} \min_{x \in \text{dom}(f)} & f(x) \\ \text{subj. to} & g_i(x) \leq 0 \quad i=1, \dots, m \\ & h_i(x) = 0 \quad i=1, \dots, p \end{aligned}$$



- Objective function $f: \text{dom}(f) \rightarrow \mathbb{R}$
- Optimization variables $x := [x_1, x_2, \dots, x_n]$
- Domain $\text{dom}(f) \subseteq \mathbb{R}^n$ of objective fcn
- inequality constraint functions (optional)
 $g_i: \mathbb{R}^n \rightarrow \mathbb{R}$, for $i=1, \dots, m$
- equality constraint functions (optional)
 $h_i: \mathbb{R}^n \rightarrow \mathbb{R}$, for $i=1, \dots, p$
- feasible set $\mathcal{X}$
 $:= \{x \in \text{dom}(f) \mid g_i(x) \leq 0, i=1, \dots, m, h_i(x) = 0, i=1, \dots, p\}$

Terminology

Now come a lot of definitions...

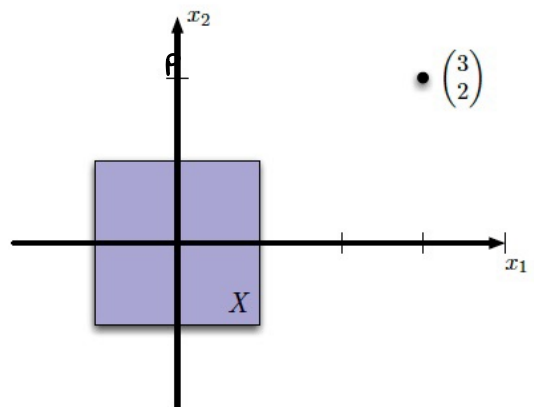
- **Feasible point**: $x \in \text{dom}(f)$ satisfying the (in-)equality constraints
- **Strictly feasible point**: Feasible $x \in \text{dom}(f)$ satisfying the inequality constraints **strictly** i.e. $g_i(x) < 0$ for $i=1, \dots, m$
- **Optimal value f^* or J^*** : Lowest possible cost value $p^* = f(x^*) \hat{=} \min_{x \in \mathcal{X}} f(x)$
- **Optimizer**: Any feasible x^* that achieves smallest cost p^* , i.e. $x^* \in \mathcal{X}$ with $f(x^*) \leq f(x)$ for all feasible $x \in \mathcal{X}$.
Optimizer not always unique! The set of solutions is: $\arg \min_{x \in \mathcal{X}} f(x) = \{x \in \mathcal{X} \mid f(x) = p^*\}$
- **(In-)active constraints**: The i th inequality constraint $g_i(x) \leq 0$ is active at $\bar{x}$ if $g_i(\bar{x}) = 0$. Equality constraints are always active.
- **Redundant constraint**: Doesn't change feasible set $\rightarrow$ removing a redundant constraint doesn't change solution.
Example: $\min_{x \in \mathbb{R}} f(x)$ subj. to $x \leq 1$
 $x \leq 2$ (redundant)

Simple Example

Problem: Find point in unit box closest to point $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

translate into standard form

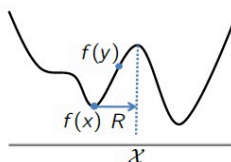
$$\begin{aligned} \min_{(x_1, x_2) \in \mathbb{R}^2} & (x_1 - 3)^2 + (x_2 - 2)^2 \\ \text{subj. to} & x_1 \leq 1 \\ & -x_1 \leq 1 \\ & x_2 \leq 1 \\ & -x_2 \leq 1 \end{aligned}$$



Optimality

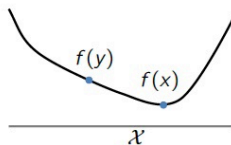
- $x \in \mathcal{X}$ is **locally optimal** if, for some $R > 0$, it satisfies

$$y \in \mathcal{X}, \|y - x\| \leq R \Rightarrow f(y) \geq f(x)$$



- $x \in \mathcal{X}$ is **globally optimal** if it satisfies

$$y \in \mathcal{X} \Rightarrow f(y) \geq f(x)$$



- If $p^* = -\infty$ the problem is **unbounded below**
- If $\mathcal{X}$ is empty, then the problem is said to be **infeasible** (convention: $p^* = \infty$)
- If $\mathcal{X} = \mathbb{R}^n$ the problem is said to be **unconstrained**

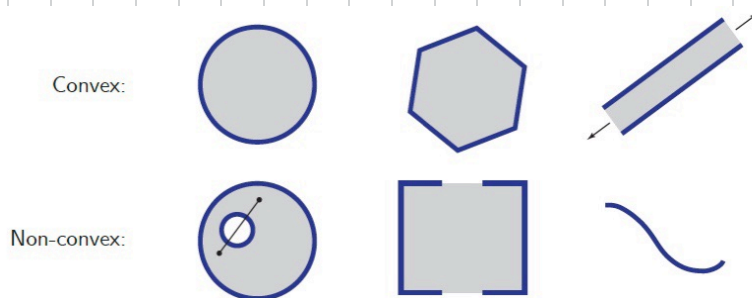
Convex Sets

To find the optimal solution, we need convex sets/functions.

They ensure that any local improvements lead directly to global optimum.

Lead to convex optimization problems, can be solved efficiently and reliably.

- Def.:** A set $\mathcal{X}$ is **convex** iff for any pair of points x and y in $\mathcal{X}$: $\lambda x + (1-\lambda)y \in \mathcal{X}, \forall \lambda \in [0,1], \forall x, y \in \mathcal{X}$
- Interpretation:** All straight line segments starting and ending in $\mathcal{X}$ stay within $\mathcal{X}$.



- Convex combination** of $x_1, \dots, x_k$: Any point x of the form $x = \theta_1 x_1 + \dots + \theta_k x_k$ with $\theta_1 + \dots + \theta_k = 1, \theta_i \geq 0$

Think of convex sets as building blocks, we need the following tools to carve them.

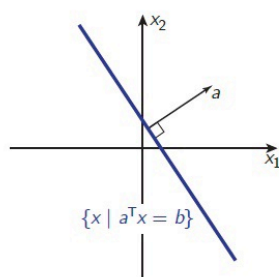
- Hyperplane:** Defined by $\{x \in \mathbb{R}^n \mid a^T x = b\}$ for $a \neq 0$, where $a \in \mathbb{R}^n$ is the normal vector to hyperplane
- Halfspace:** Everything on one side of hyperplane $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$ for $a \neq 0$. It can either be **open** (strict inequality) or **closed** (non-strict equality).

For $n=2$, hyperplanes define lines.

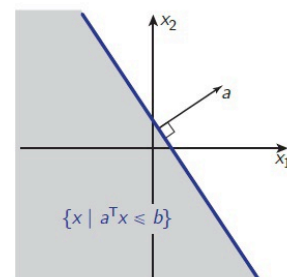
affine & convex

For $n=3$ hyperplanes define planes.

just convex



A hyperplane



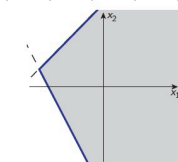
A closed halfspace

We can use hyperplanes & halfspaces to build:

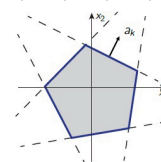
- Polyhedron**: Intersection of finite numbers of closed halfspaces:

$$P = \{x \mid a_i^T x \leq b_i, i=1, \dots, n\} = \{x \mid Ax \leq b\}$$

where $A := [a_1, a_2, \dots, a_m]^T$ and $b := [b_1, b_2, \dots, b_m]^T$



An (unbounded) polyhedron



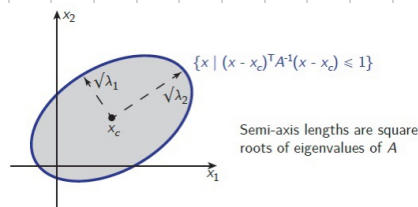
A polytope



- Polytope**: **bounded** polyhedron

- Ellipsoids**: Set defined as $\{x \mid (x - x_c)^T A^{-1} (x - x_c) \leq 1\}$

centre of ellipsoid
 $A \succ 0$ (pos. def.)



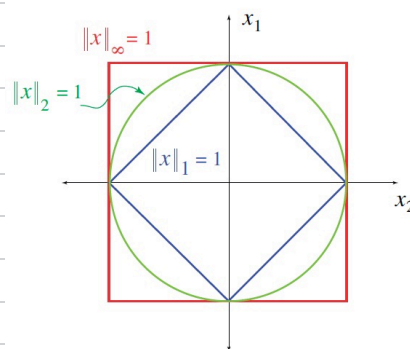
$\hookrightarrow$ if $A = r^2 I \rightarrow$ Euclidean ball $B(x_c, r) := \{x \mid \|x - x_c\|_2 \leq r\}$

- norm-ball**: defined by $\{x \mid \|x - x_c\| \leq r\}$ $\leftarrow$ always convex for any norm

ball centre $\leftarrow$ radius

most common ℓ_p norms are:

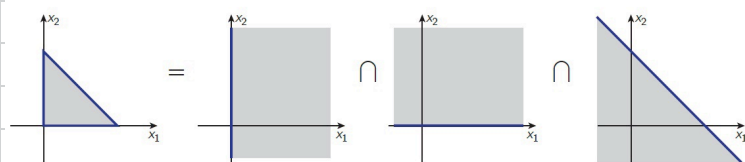
- $p=2$ (Euclidean norm): $\|x\|_2 = \sqrt{\sum_i x_i^2}$
- $p=1$ (sum of all absolutes): $\|x\|_1 = \sum_i |x_i|$
- $p=\infty$ (Largest absolute value): $\|x\|_\infty = \max_i |x_i|$



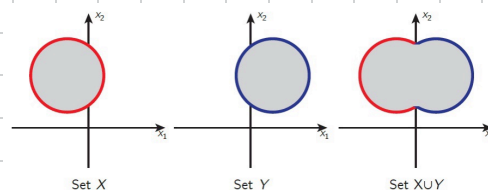
Now we have seen many different (simple) convex sets. We can manipulate them to form more complex sets. In particular we can use the following operation:

- Intersection** of two or more convex sets is itself convex.

Any convex set can be written as (possible infinite) intersection of halfplanes.



- Note**: **union** of two sets is **not** convex in general, even if original sets were convex.



For a convex optimization problem we don't only need convex sets but also convex functions.

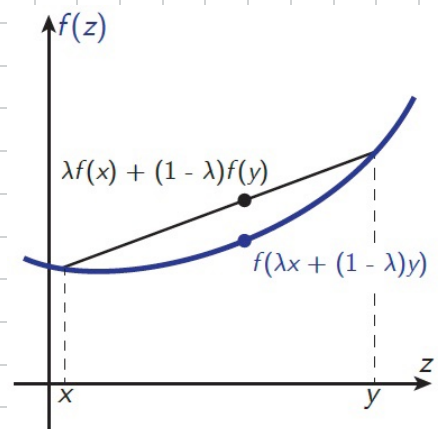
Convex Functions

- Def.: A function $f: \text{dom}(f) \rightarrow \mathbb{R}$ is **convex** iff $\text{dom}(f)$ is convex **and**

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y), \quad \forall \lambda \in (0,1), \quad \forall x, y \in \text{dom}(f)$$

graphically:

The function $f: \text{dom}(f) \rightarrow \mathbb{R}$ is **strictly convex** if this inequality is strict.



- Side note: f is **concave** iff $\text{dom}(f)$ is convex & $-f$ is convex.

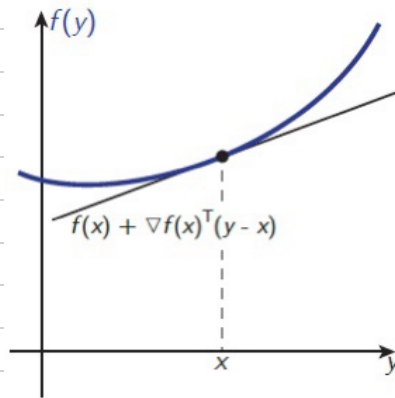
Instead of picking every possible pair of points and running the previous check, you can use calculus and check one of the following two conditions:

1st order Condition for Convexity:

Def.: A differentiable function $f: \text{dom}(f) \rightarrow \mathbb{R}$ with a convex domain is convex iff

$$f(y) \geq f(x) + \nabla f(x)^T (y-x), \quad \forall x, y \in \text{dom}(f) \quad \leftarrow \text{gradient is given by } \nabla f(x) = \left[\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right]^T$$

→ This essentially means first-order approximation of f around any point is a **global underestimator** of f .



2nd order Condition for Convexity

Def.: A twice differentiable function $f: \text{dom}(f) \rightarrow \mathbb{R}$ with convex domain $\text{dom}(f)$ is convex iff

$$\nabla^2 f(x) \succeq 0 \quad \forall x \in \text{dom}(f) \quad \leftarrow \text{Hessian } \nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

↑ semi positive definite

Note: If $\text{dom}(f)$ is convex and $\nabla^2 f(x) \succ 0$ for all $x \in \text{dom}(f)$, then $f(x)$ is **strictly convex**

Now that we have seen how convex functions are defined, here are some examples:

Convex functions:

- Affine: $ax + b$ for any $a, b \in \mathbb{R}$
- Exponential: e^{ax} for any $a \in \mathbb{R}$
- Powers: x^α on domain $\mathbb{R}_{++}$, for $\alpha \geq 1$ or $\alpha \leq 0$
- Vector norms on $\mathbb{R}^n$: $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$, for $p \geq 1$, $\|x\|_\infty = \max_i |x_i|$

Concave functions:

- Affine: $ax + b$ for any $a, b \in \mathbb{R}$
- Powers: x^α on domain $\mathbb{R}_{++}$, for $0 \leq \alpha \leq 1$
- Logarithm: $\log x$ on domain $\mathbb{R}_{++}$
- Entropy: $-x \log x$ on domain $\mathbb{R}_{++}$

Certain operations preserve the convexity of functions:

- Non-negative weighted sum
- Composition with affine function
↳ $f(Ax+b)$ convex if f is convex

and many other possibilities...

- Pointwise maximum and supremum

- Partial minimization

↳ $f(x) = \max \{f_1(x), f_2(x)\}$ convex if f_1, f_2 convex

We have seen how convex sets & functions are defined, now we'll set up an important class of optimization problem...

Convex Optimization Problem

A **convex** optimization problem in standard form:

$$\begin{array}{ll} \min_{x \in \text{dom}(f)} f(x) \\ \text{subi. to } g_i(x) \leq 0 & i=1, \dots, m \\ a_i^T x = b_i & i=1, \dots, p \end{array}$$

gather affine
constraints into matrix form

$$\begin{array}{ll} \min_{x \in \text{dom}(f)} f(x) \\ \text{subi. to } g_i(x) \leq 0 & i=1, \dots, m \\ Ax = b & A \in \mathbb{R}^{p \times m} \end{array}$$

- $f, g_1, \dots, g_m$ are convex functions
- $\text{dom}(f)$ is a convex set
- equality constraint functions $h_i(x) = a_i^T x - b_i$ are all affine

Important properties:

- Feasible set of a convex optimization problem is convex
- Any **locally optimal solution** is **globally optimal**

Equivalent Optimization Problems

Sometimes a problem is messy to solve in original form, we use tricks to rewrite in "standard form" that optimization software prefers.

- Introducing equality constraints

$$\begin{array}{ll} \min_x f(A_0 x + b_0) \\ \text{subi. to } g_i(A_i x + b_i) \leq 0 & i=1, \dots, m \end{array}$$

← equivalent →

$$\begin{array}{ll} \min_{x, y_i} f(y_0) \\ \text{subi. to } g_i(y_i) \leq 0 & i=1, \dots, m \\ Ax_i + b_i = y_i & i=0, 1, \dots, m \end{array}$$

↑ complicated expression inside $f(\cdot)$

- Introducing slack variables for linear inequalities

$$\begin{array}{ll} \min_x f(x) \\ \text{subi. to } A_i x \leq b_i & i=1, \dots, m \end{array}$$

← equivalent →

$$\begin{array}{ll} \min_{x, s_i} f(x) \\ \text{subi. to } A_i x + s_i = b_i & i=1, \dots, m \\ s_i \geq 0 & i=1, \dots, m \end{array}$$

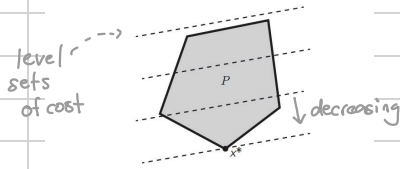
↑
some solver can handle
equality constraints better

Linear Programs

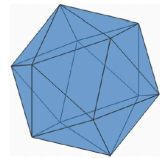
Linear programs are a special type of convex optimization problems. They have the following form:
They have **affine** cost/constraint functions:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subi. to} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

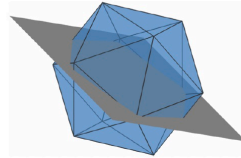
- Feasible set P is a polyhedron
- If P is empty, the problem is infeasible



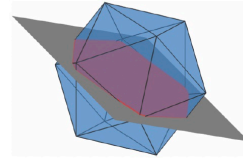
Linear optimization on a polytope.



(a) $Gx \leq h$



(b) $A_j^T x = b_j$



(c) $Gx \leq h \cap A_j^T x = b_j$

Graphical interpretation, denote

p^* : optimal value

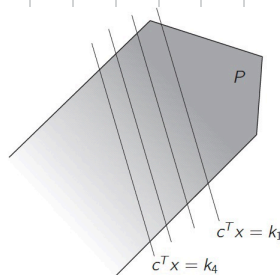
X_{opt} : set of optimizers

We distinguish between three cases:

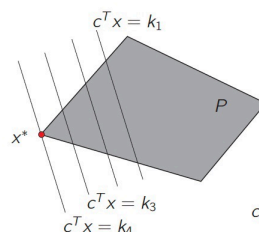
Case 1: The LP solution is unbounded $\rightarrow p^* = -\infty$

Case 2: The LP solution is bounded $\rightarrow p^* > -\infty$ and optimizer unique. X_{opt} is singleton

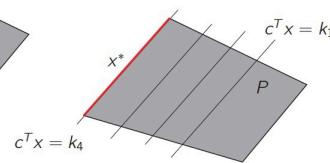
Case 3: The LP solution is bounded and there are multiple optima. $\rightarrow X_{\text{opt}}$ is subset of $\mathbb{R}^s$ (can be bounded/unbounded)



(a) Case 1



(b) Case 2



(c) Case 3

Quadratic Programs

Another special type of convex optimization problem

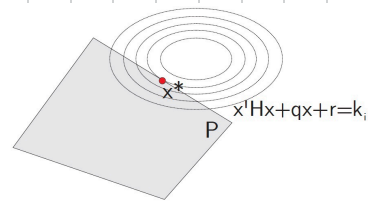
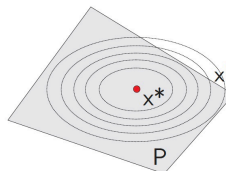
$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \frac{1}{2} x^T H x + q^T x + r \\ \text{subi. to} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

- Constant r can be left out (no effect on solution)
- Convex if $H \succeq 0$
- Problems with concave objective $H \preceq 0$ are QPs, but hard

We distinguish between two cases:

Case 1: The optimizer lies strictly inside feasible polyhedron

Case 2: The optimizer lies on the boundary of feasible polyhedron.



We have now seen how the most important optimization problems look like, let's return to our main objective...

Constrained Infinite Time Optimal Control

← the problem we actually want to solve

$$J_{\infty}^*(x(0)) = \min_{u(\cdot)} \sum_{i=0}^{\infty} l(x_i, u_i)$$

$$\begin{aligned} \text{subi. to } & x_{i+1} = Ax_i + Bu_i, i=0, \dots, \infty \\ & x_i \in X, u_i \in U, i=0, \dots, \infty \\ & x_0 = x(0) \end{aligned}$$

- Stage cost $l(x, u)$: "cost" of being in x and applying u
- Optimizing over a trajectory provides **tradeoff between short- and long-term benefits** of actions
- This control law has many beneficial properties, but can't compute, because of infinite number of variables

That's why we need...

Constrained Finite Time Optimal Control

$$J^*(x(k)) = \min_U l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i)$$

$$\begin{aligned} \text{subi. to } & x_{i+1} = Ax_i + Bu_i, i=0, \dots, N-1 \\ & x_i \in X, u_i \in U, i=0, \dots, N-1 \end{aligned}$$

$$x_N \in X_f$$

$$x(0) = x(k)$$

$$\text{where } U = \{u_0, \dots, u_{N-1}\}$$

Truncate after finite horizon:

- N : time horizon
- $l_f(x_N)$: Approximates 'tail' of cost
- X_f : Approximates 'tail' of constraints

We can even further simplify this, remember how we used a quadratic cost for LQR?

Constrained Linear Optimal Control

We now consider the following constrained optimal control problem with quadratic cost & linear dynamics.

$$J^*(x(k)) = \min_U x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i$$

$$\begin{aligned} \text{subi. to } & x_{i+1} = Ax_i + Bu_i, i=0, \dots, N-1 \\ & x_i \in X, u_i \in U, i=0, \dots, N-1 \\ & x_N \in X_f \\ & x_0 = x(k) \end{aligned}$$

- N : time horizon

- X, U, X_f : polyhedral sets

- $P \succeq 0, Q \succeq 0, R \succ 0$

↑ compute via ARE or Lyapunov

↑ Note: possible to consider different costs,

e.g. $l_f(x_N) = \|x_N\|_p$ & $l(x_i, u_i) = \|x_i\|_p + \|u_i\|_p$ with P, Q, R full column rank matrices and $p=1$ or ∞

All of these constraints introduced above limit the starting state x_0 , we can define the **Feasible Set**

$$\mathcal{X}_N = \{x_0 \in \mathbb{R}^n \mid \exists (u_0, \dots, u_{N-1}) \text{ such that } x_i \in \mathcal{X}, u_i \in \mathcal{U}, \\ i=0, \dots, N-1, x_N \in \mathcal{X}_f, \text{ where } x_{i+1} = Ax_i + Bu_i\}$$

translated into words: this set contains all of the initial states, for which there exist control inputs over the horizon, s.t. $x_i \in \mathcal{X}$, $u_i \in \mathcal{U}$ fulfill system dynamics and x_N ends up in a nice final state (so it can fulfill constraints just as good in the future)

Reminder: Unconstrained Solution

no state & input constraints

Last week we solved the unconstrained problem, for this we had a nice closed-form solution:

We looked at two cases:

Case 1: time-varying linear control law $u_i^* = F_i x_i$, $i=0, \dots, N-1$

Case 2: $N \rightarrow \infty$, i.e. time-invariant control law $u_i^* = F_\infty x_i$, $i=0, 1, \dots$

Next let's look at how to compute the finite time constrained optimal controllers...

For this we need to transform our known Quadratic Cost CFTOC into QP

$J^*(x(k)) = \min_u x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i$ subj. to $x_{i+1} = Ax_i + Bu_i, \quad i=0, \dots, N-1$ $x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}, \quad i=0, \dots, N-1$ $x_N \in \mathcal{X}_f$ $x_0 = x(k)$	transform →	$\min_{z \in \mathbb{R}^n} \frac{1}{2} z^T H z + q^T z + r$ subj. to $Gz \leq h$ $Az = b$
---	---	--

There are two ways to do this...

The 1st way...

Construction of QP with substitution

The main idea is to substitute the state equations, now we don't have to solve for both x and u but only u by making all future states dependent on current state $x(k)$

$$x_{i+1} = Ax_i + Bu_i$$

$$x_0 = x(k)$$

First we can rewrite the cost as:

$$J(x(k), u) = U^T H U + 2x(k)^T F U + x(k)^T Y x(k)$$

$$= \begin{bmatrix} U^T & x(k)^T \end{bmatrix} \begin{bmatrix} H & F^T \\ F & Y \end{bmatrix} \begin{bmatrix} U^T & x(k)^T \end{bmatrix}^T$$

$$\cdot H := (S^u)^T \bar{Q} S^u + \bar{R}$$

$$\cdot F := (S^x)^T \bar{Q} S^u$$

$$\cdot Y := (S^x)^T \bar{Q} S^x$$

$$\text{and: } X := S^x x(0) + S^u U$$

$$\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{bmatrix} = \underbrace{\begin{bmatrix} I \\ A \\ \vdots \\ A^N \end{bmatrix}}_{S^x} x(0) + \underbrace{\begin{bmatrix} 0 & \dots & 0 \\ B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ A^{N-1}B & \dots & AB & B \end{bmatrix}}_{S^u} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}$$

$$\bar{Q} = \text{blockdiag}(Q, Q, P)$$

$$\bar{R} = \text{blockdiag}(R, \dots, R)$$

Second rewrite constraints compactly, remember we are constraining U

$$G U \leq w + E x(k)$$

These G, w, E for $\mathcal{X} = \{x | A_x x \leq b_x\}$, $\mathcal{U} = \{u | A_u u \leq b_u\}$ $\mathcal{X}_f = \{x | A_f x \leq b_f\}$ are given by:

$$G = \begin{bmatrix} A_u & 0 & \dots & 0 \\ 0 & A_u & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_u \\ 0 & 0 & \dots & 0 \\ A_x B & 0 & \dots & 0 \\ A_x A B & A_x B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_f A^{N-1} B & A_f A^{N-2} B & \dots & A_f B \end{bmatrix}, E = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ -A_x \\ -A_x A \\ -A_x A^2 \\ \vdots \\ -A_f A^N \end{bmatrix}, w = \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \\ b_x \\ b_x \\ b_x \\ \vdots \\ b_f \end{bmatrix}$$

Putting it all together

$$J^*(x(k)) = \min_U \begin{bmatrix} U^T & x(k)^T \end{bmatrix} \begin{bmatrix} H & F^T \\ F & Y \end{bmatrix} \begin{bmatrix} U^T & x(k)^T \end{bmatrix}^T$$

subj. to $G U \leq w + E x(k)$

For given $x(k)$, U^* can be found via QP solver

The 2nd way...

Construction of QP without substitution

We can also keep state equations as equality constraints and stack all our optimization

variables in one vector, our optimization problem now looks like:

$$\begin{aligned} J^*(x(k)) = \min_z & \begin{bmatrix} z^T & x(k)^T \end{bmatrix} \begin{bmatrix} \bar{H} & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} z^T & x(k)^T \end{bmatrix}^T \\ \text{subj. to } & G_{in} z \leq W_{in} + E_{in} x(k) \\ & G_{eq} z = E_{eq} x(k) \end{aligned}$$

Let's break it down:

- Our stacked optimization variable:

$$z = [x_1^T \dots x_N^T \ u_0^T \dots \ u_{N-1}^T]$$

- We can find G_{eq}, E_{eq} from system dynamics:

$$G_{eq} = \begin{bmatrix} I & & & & & \\ -A & I & & & & \\ & -A & I & & & \\ & & \ddots & \ddots & & \\ & & & -A & I & \\ & & & & & -B \\ & & & & & -B \\ & & & & & & \ddots \\ & & & & & & & -B \end{bmatrix}, E_{eq} = \begin{bmatrix} A \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- We can find G_{in}, W_{in}, E_{in} by using the following constraints for each time step:

$$X = \{x | A_x x \leq b_x\}, U = \{u | A_u u \leq b_u\} \quad X_f = \{x | A_f x \leq b_f\}$$

$$G_{in} = \begin{bmatrix} 0 & & & 0 & & \\ A_x & & & 0 & & \\ & A_x & & & 0 & \\ & & \ddots & & & 0 \\ & & & A_f & & 0 \\ 0 & & & & A_u & \\ & 0 & & & A_u & \\ & & \ddots & & & A_u \\ & & & 0 & & A_u \\ & & & & & 0 \end{bmatrix}, W_{in} = \begin{bmatrix} b_x \\ b_x \\ b_x \\ \vdots \\ b_x \\ b_f \\ b_u \\ b_u \\ \vdots \\ b_u \\ b_u \end{bmatrix}$$

$$E_{in} = [-A_x^T \ 0 \ \dots \ 0]^T$$

- The H matrix can be found by building cost function from MPC $x_N^T P x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i$

$$\bar{H} = \begin{bmatrix} Q & & & & \\ & \ddots & & & \\ & & Q & & \\ & & & P & \\ & & & & R \\ & & & & & \ddots \\ & & & & & & R \end{bmatrix}$$

Exercise 1

Which of the following sets are convex?

- $P_1 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$
- $P_2 = \{x \in \mathbb{R}^n \mid x^T H x \leq 2, H \succ 0\}$
- $P_3 = \{x \in \mathbb{R}^n \mid x^T x = 0\}$
- $P_4 = \{x \in \mathbb{R}^n \mid x^T x = 2\}$

Solution:

P_1 convex: Polyhedra sets are convex

P_2 convex: By 2nd order condition $\nabla^2 f(x) = H$, which is pos. def.

P_3 convex: Set is given by $\{0\}$ by def.: $1 \cdot 0 + (1-1) \cdot 0 = 0$

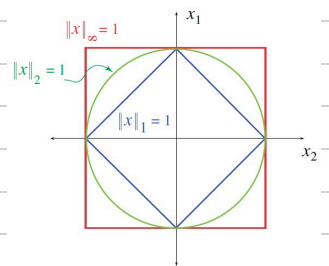
P_4 non-convex: Set is given by $\{-\sqrt{2}, \sqrt{2}\}$ by def. with $\lambda = 0.5$: $0.5 \cdot (\sqrt{2}) + 0.5 \cdot (-\sqrt{2}) = 0 \notin P_4$

Exercise 2

Consider the optimization problem

$$P: \begin{aligned} x_p^* &= \operatorname{argmin}_{x \in \mathbb{R}^2} \|x\|_p \\ \text{s.t. } & x_1 + x_2 = 3 \\ & (x_2 - 2)^2 \leq 1 \end{aligned}$$

With $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$



i) Draw feasible set P

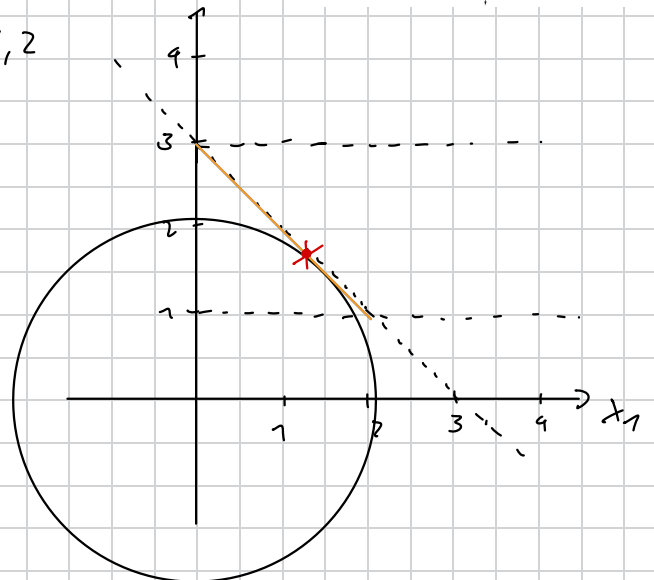
ii) State the set of all optimal solution x_1^* for $p=1, 2$

Solution: $|x_2 - 2| \leq 1$

$$\begin{aligned} \leadsto x_2 - 2 &\leq 1 \leadsto x_2 \leq 3 \\ -x_2 + 2 &\leq 1 \leadsto x_2 \geq 1 \end{aligned}$$

• 1 norm: $x_1^* = \{x \mid x_1 + x_2 = 3, (x_2 - 2)^2 \leq 1\}$

• 2-norm: $x_2^* = \{x = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}\}$



Summary

Today we looked at...

- How to construct convex optimization problems including:
 - Convex sets
 - Convex cost functions
 - Convex inequality constraints
 - Affine equality constraints
- Benefits of convex problems: Local = Global optimality $\leadsto$ only need to find one optimum
- Convex optimization problems can be solved efficiently
- CFTOC w/linear dynamics, quadratic cost, polytopic constraints $\longrightarrow$ QP